

SECTION 1: RECTANGULAR SUNK KEY DESIGN

System Parameters: Shaft diameter $d = 50\text{ mm}$, Power $P = 25\text{ kW}$, Speed $N = 250\text{ rpm}$, allowable key shear stress $\tau = 40\text{ MPa}$, allowable key crushing stress $\sigma_c = 80\text{ MPa}$, standard key proportions from IS: 2048-1962.

Design Protocol: Calculate torque, select standard key cross-section from table, evaluate required key length in shear and crushing, and adopt the governing safe length.

1. Torque transmitted by shaft	$T = \frac{60P}{2\pi N}$ $= \frac{60 \times 25 \times 10^3}{2\pi \times 250}$ $= 954.93\text{ N}\cdot\text{m}$ $= 9.5493 \times 10^5\text{ N}\cdot\text{mm}$
2. Standard key cross-section selection	$d = 50\text{ mm}$ $w = 16\text{ mm}$ $t = 10\text{ mm}$
3. Required key length from shear criterion	$T = w \cdot l \cdot \tau \cdot \frac{d}{2}$ $9.5493 \times 10^5 = 16 \times l \times 40 \times 25$ $l = \frac{9.5493 \times 10^5}{16000}$ $= 59.7\text{ mm}$
4. Required key length from crushing criterion	$T = \left(\frac{t}{2}\right) \cdot l \cdot \sigma_c \cdot \frac{d}{2}$ $9.5493 \times 10^5 = 5 \times l \times 80 \times 25$ $l = \frac{9.5493 \times 10^5}{10000}$ $= 95.5\text{ mm}$
5. Adopted key length	$l_{\text{shear}} = 59.7\text{ mm}$ $l_{\text{crushing}} = 95.5\text{ mm}$ $l = 95.5\text{ mm}$ $\Rightarrow l = 96\text{ mm}$
6. Design output	Rectangular sunk key dimensions: width $w = 16\text{ mm}$, thickness $t = 10\text{ mm}$, adopted length $l = 96\text{ mm}$.

SECTION 2: SQUARE SUNK KEY DESIGN

System Parameters: Shaft diameter $d = 45\text{ mm}$, Power $P = 20\text{ kW}$, Speed $N = 200\text{ rpm}$, allowable key shear stress $\tau = 45\text{ MPa}$, allowable crushing stress $\sigma_c = 90\text{ MPa}$.

Design Protocol: Compute torque, select square key size where $w = t$, then size key length from the governing strength relation.

1. Torque transmitted by shaft

$$\begin{aligned} T &= \frac{60P}{2\pi N} \\ &= \frac{60 \times 20 \times 10^3}{2\pi \times 200} \\ &= 954.93\text{ N}\cdot\text{m} \\ &= 9.5493 \times 10^5\text{ N}\cdot\text{mm} \end{aligned}$$

2. Selected square key cross-section

$$\begin{aligned} d &= 45\text{ mm} \\ w &= 16\text{ mm} \\ t &= 16\text{ mm} \end{aligned}$$

3. Required key length

$$\begin{aligned} T &= w \cdot l \cdot \tau \cdot \frac{d}{2} \\ 9.5493 \times 10^5 &= 16 \times l \times 45 \times 22.5 \\ l &= \frac{9.5493 \times 10^5}{16200} \\ &= 58.9\text{ mm} \\ \Rightarrow l &= 60\text{ mm} \end{aligned}$$

4. Design output

Square sunk key dimensions: width $w = 16\text{ mm}$, thickness $t = 16\text{ mm}$, adopted length $l = 60\text{ mm}$.

SECTION 3: KEYWAY EFFECT ON SHAFT STRENGTH

System Parameters: Shaft diameter $d = 50$ mm, keyway width $w = 16$ mm, keyway depth $h = 5$ mm.

Design Protocol: Use empirical keyway reduction factor and compute strength ratio of keyed shaft relative to a solid shaft.

1. Keyway weakening factor

$$\begin{aligned} e &= 1 - 0.2\left(\frac{w}{d}\right) - 1.1\left(\frac{h}{d}\right) \\ &= 1 - 0.2\left(\frac{16}{50}\right) - 1.1\left(\frac{5}{50}\right) \\ &= 1 - 0.064 - 0.11 \\ &= 0.826 \end{aligned}$$

2. Strength ratio of keyed shaft to solid shaft

$$\begin{aligned} \frac{T_{\text{keyway}}}{T_{\text{solid}}} &= e \\ &= 0.826 \\ &= 82.6\% \end{aligned}$$

3. Design output

The keyed shaft retains 82.6% of the torsional strength of an equivalent solid shaft.

SECTION REF 1: TABLE 13.1 — STANDARD PARALLEL, TAPERED, AND GIB
HEAD KEY PROPORTIONS

Shaft d up to (mm)	Key w (mm)	Key t (mm)	Shaft d up to (mm)	Key w (mm)	Key t (mm)
6	2	2	85	25	14
8	3	3	95	28	16
10	4	4	110	32	18
12	5	5	130	36	20
17	6	6	150	40	22
22	8	7	170	45	25
30	10	8	200	50	28
38	12	8	230	56	32
44	14	9	260	63	32
50	16	10	290	70	36
58	18	11	330	80	40
65	20	12	380	90	45
75	22	14	440	100	50

Reference: IS: 2048-1962 standard key dimensions used for preliminary key cross-section selection.